GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 2

COURSE: ENGINEERING MATHEMATICS - II (ETMA-203)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETMA-203

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Find the integrating factor for the linear differential equation dy/dx + P(x)y = Q(x).

Q1. (b) State Cauchy-Riemann equations in polar and Cartesian coordinates.

Q1. (c) Define harmonic function and write condition for harmonic conjugacy.

Q1. (d) State D'Alembert Ratio Test for convergence of infinite series.

Q1. (e) State Cauchy Integral Formula for analytic functions.

Q1. (f) Find residues of f(z) = 1 / (z^2 + 1) at its poles.

Q1. (g) Evaluate the Beta function B(m, n) in terms of Gamma functions.

Q1. (h) Define order and degree of a differential equation with an example.

Q1. (i) State Cauchy Residue Theorem for contour integration.

Q1. (j) Explain radius of convergence of a power series.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Solve the differential equation (D^2 - 4D + 4)y = 8(x^2 + e^(2x) + sin(2x)).

Q2. (b) Using method of variation of parameters, solve (D^2 + 1)y = sec(x).

Q2. (c) Determine whether the function u = e^x(x cos y - y sin y) is harmonic and find its conjugate v.

Q2. (d) Evaluate integral of (z^2 + 1) / (z - 1)(z - 2) around circle |z| = 3 using Cauchy Residue Theorem.

Q2. (e) Test the convergence of the series sum from n=1 to inf of ((n+1)/n)^(n^2).

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Solve Cauchy-Euler homogeneous differential equation x^2 y'' - 2x y' - 4y = x^2 + 2 log(x). [10 Marks]

-- OR --

(b) Solve simultaneous differential equations dx/dt + 2y = -sin(t), dy/dt - 2x = cos(t). [10 Marks]

Q4. (a) Derive Milne-Thomson method and find analytic function f(z) whose real part is u = x^3 - 3xy^2. [10 Marks]

-- OR --

(b) Find the bilinear transformation that maps points z = (1, i, -1) to w = (0, 1, inf). [10 Marks]

Q5. (a) Expand f(z) = 1 / ((z - 1)(z - 2)) in Laurent series in regions 1 < |z| < 2 and |z| > 2. [10 Marks]

-- OR --

(b) Evaluate real definite integral from 0 to 2pi of d_theta / (5 + 4 cos(theta)) using contour integration. [10 Marks]

Q6. (a) Prove that integral from -inf to +inf of cos(mx) / (x^2 + a^2) dx = (pi/a) e^(-ma) for a > 0. [10 Marks]

-- OR --

(b) Solve by Frobenius method: x^2 y'' + x y' + (x^2 - 4)y = 0 near x = 0. [10 Marks]

Q7. (a) Prove duplication formula for Gamma functions: Gamma(m) Gamma(m + 1/2) = sqrt(pi) 2^(1-2m) Gamma(2m). [1

-- OR --

(b) Test convergence and absolute convergence of alternating series sum (-1)^n / (n log n). [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****